

Rauf Shah, MS, PhD, MPH-EPI (Expected 2026)

Lead Health Data Coordinator II | Healthcare | Biomedical Informatics Scientist

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[Google Scholar](#) | [Website](#)

Profile Summary

Healthcare informatics and biomedical Scientist with a rich blend of experimental and digital expertise. Specializing in designing and validating liquid biopsy assays (cfDNA/ctDNA, ddPCR, NGS) and integrating multi-omics and genomic datasets to uncover actionable biomarkers in hematologic malignancies, I bring transformative insights from bench to bedside. Adept at steering cross-functional teams, I have led initiatives implementing cloud-native data platforms that seamlessly integrate EHRs, patient registries, and dynamic dashboards for real-time decision-support. My work, consistently featured in high-impact publications, bridges scientific discovery with enhanced clinical outcomes.

Accreditation-aligned leadership: I design accreditation-ready systems, governance frameworks, SOPs, audit trails, evidence dossiers, KPI dashboards, and executive commentaries, linking institutional strategy to measurable outcomes. I operate directly under the Division Chief, partner with the Cancer Center Director, and occasionally interact with the Vice Chancellor on project-specific briefings. I lead self-study inputs, curriculum evaluation, and continuous improvement activities; design and implement IRB protocols; and mentor analysts, clinicians, and fellows on documentation and reporting quality.

Key Skills

- NGS & High-Throughput Data Analysis
- Cancer Genomics & Liquid Biopsy
- Metabolic Modeling & Genomic Epidemiology
- Data Architecture & Machine Learning
- Cloud Computing (AWS) & HPC
- Cross-Functional Leadership & Collaboration
- Multicenter data governance
- Grant writing & budgets
- Mentoring & teaching
- Accreditation & Quality Systems (self-study, SOPs, audit readiness, evidence libraries)
- Curriculum Governance & Evaluation (policies, CE oversight, outcomes mapping)
- Executive Reporting (KPI dashboards, outcomes/equity metrics, English commentaries)
- Stakeholder Engagement & Consensus-Building (faculty, registry, IRB, IT, clinical)
- IRB/Research Oversight (protocol design/implementation; compliance & documentation)

Technical Skills

- Operating Systems: Linux, macOS, Windows
- Programming Languages: Python, R, MATLAB, Java, JavaScript, Bash
- Data Analysis & Multi-Omics: RNA-seq, metabolomics, NGS variant calling, clonal evolution, metabolic model integration
- NGS & Bioinformatics Tools: GATK, STAR, Salmon, MACS2, cnvkit, DESeq2
- Statistical Packages: R (various packages), Python (NumPy, SciPy, Pandas), SPSS
- Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, Keras, LightGBM, AutoML
- Deep Learning Applications: CNNs for image analysis, LSTMs for time-series modeling, Transformer models for genomics and NLP
- Data Science & Visualization: Jupyter, Matplotlib, Seaborn, ggplot2, Tableau, Power BI, SQL
- Wet Lab Techniques: cfDNA isolation (MagMAX, QIAamp), ddPCR (Bio-Rad QX200), qPCR, Illumina library prep, Bioanalyzer, TapeStation, cell culture
- Tools & Platforms: Git, VS Code, Docker, AWS (S3, EC2, Athena), Google Cloud, OriginLab, CI/CD
- Standards & Interoperability: FHIR/HAPI, NAACCR, ICD/ICD-O; Identity & Access: Keycloak
- Workflow & Reproducibility: Docker Compose, GitHub Actions; Governance: role-based access, change control

Education

- **MPH-Epidemiology** (Expected 2026), University of Nebraska Medical Center, Omaha, NE, 2024-Present
- **PHD., Bioinformatics**, Pondicherry University, India, 2018
- **MS., Bioinformatics**, Jamia Millia Islamia, Delhi, India, 2010

- BSc., University of Kashmir, Srinagar, India, 2006

Certifications

- *Foundations of Business Intelligence*, 2025, (Google):
<https://www.coursera.org/account/accomplishments/verify/QXIOQBSLG363>
- *Decisions, Decisions: Dashboards and Reports*, (Google): 2025,
<https://www.coursera.org/account/accomplishments/verify/APQRC2PR7YLI>
- *The Path to Insights: Data Models and Pipelines*, 2025 - (Google):
<https://www.coursera.org/account/accomplishments/verify/EQ6RUR5WC43S>
- *Google Business Intelligence, Professional Certificate*: 2025,
<https://www.coursera.org/account/accomplishments/professional-cert/WMZUZFR3N5GH>
- Phlebotomy Technician, Nebraska (#NE-8367-70958), 2022 - Supports liquid biopsy workflows

Professional Experience

Lead Health Data Coordinator II, UNMC, Internal Medicine, Lymphoma Study Group (Dec 2024-Present)

- Spearhead data management for multicenter lymphoma studies with over (> 32k patients), ensuring 100% FDA compliance.
- Lymphoma Clinical data and Registry maintenance, Admin and Manager
- Engineered an AWS architecture that cuts data latency by 40%, accelerating research insights and publications.
- Collaborate with clinicians, geneticists, and bioinformaticians to integrate genomic and clinical and Biomedical health-care data.
- Develop ML-powered dashboards and frameworks for EHR, digital pathology, and patient registries.
- Establish accreditation-ready governance: policies, SOPs, audit trails, evidence libraries, KPI dashboards, and executive commentaries.
- Operate directly under the Division Chief; partner with the Cancer Center Director; brief the Vice Chancellor on project-specific progress.
- Implement standards-based FHIR/HAPI data services with validation checks, versioned transformations, and role-based access controls.
- Support clinicians and medical fellows with data queries and clinical research project design; lead IRB documentation and submissions.

Lead Computational Biologist, UNMC, Pathology & Microbiology (Nov 2022-Dec 2024)

- Successfully completed project Towards Digitalization of Clinical, Molecular, Digital Pathology & Cancer Genomic data. I am the developer and admin of these Apps at UNMC.
- The project is hosted at AWS & can be accessed via Combined clinical and Digital Pathology Portals:
 - <https://hemepath.unmc.edu> (Developer and Admin)
 - <https://pathology.unmc.edu/apps/table.html> (Developer and Admin)
- Established liquid biopsy SOPs and optimized plasma cf-DNA preservation.
- Led project to digitalize clinical, molecular, and pathology data on AWS with specialized portals.
- Developed targeted NGS panel (32 genes) for peripheral T-cell lymphoma liquid biopsy monitoring.
- Optimized ddPCR assays detecting hotspot mutations at 0.01 % VAF; validated on 120 plasma samples.
- Integrated RNA-seq, WES, and digital pathology data to identify subtype-specific vulnerabilities (ASH 2024 abstract).
- Mentored 2 graduate students in genomic workflows and wet-lab validation
- Built quality systems (SOPs, validation plans, QC dashboards, documentation repositories) to support external review and audit readiness.

Data Scientist| UNMC, Pathology & Microbiology (Feb 2022-Nov 2022)

- Conducted NGS data analysis and liquid biopsy research in cancer genomics.

Research Scientist, UNMC, Pathology & Microbiology (Dec 2021-Feb 2022)

- Engaged in Cancer Genomics projects with a focus on NGS data analysis.

Postdoctoral Research Associate, University of Nebraska-Lincoln (Nov 2017-Nov 2021)

- Reconstructed and analyzed mixed-bag metabolic models of soil Viral-AMGs-bacteria-fungi.
- Developed agent-based immune system models and metabolic models of CD4+ T cells.
- Contributed to consortium grant proposals submitted to the US DOE Office of Science.

Senior Research Fellow, International Centre for Genetic Engineering and Biotechnology (ICGEB), Delhi & Pondicherry University (2014-2017)

- Led marine microalgae biofuel project; scaled Nannochloropsis cultures to 300 L photobioreactors.
- Published first GSM of Nannochloropsis gaditana (iRJ1321) in Algal Research.
- Taught workshops on qPCR and metabolic modeling to postgraduate students.

Junior Research Fellow, Pondicherry University (2012-2014)

- Conducted research on culturing and characterizing marine algal strains.
- Led large-scale culturing of Nannochloropsis sp. for biofuel extraction.

Research Assistant, Institute of Genomic and Integrative Biology (2010-2012)

- Enhanced research capabilities through automated literature mining.
- Created pubmed.mineR, R package for literature mining:
<https://www.rdocumentation.org/packages/pubmed.mineR/versions/1.0.19>
- Automated curation pipeline for PubMed abstracts improving annotation throughput by 60%.

Teaching & Mentoring Experience

- **Instructor**, University of Nebraska, BIOC 435: Dynamics of Biochemical and Biological Networks (2018-2020).
- Mentored six undergraduate and graduate research projects (2015-2019) across computational systems biology and bioinformatics.

Governance, Accreditation & Committee Leadership

Elected Member, Curriculum Committee, UNMC College of Public Health (Oct 2025-Present)

- Student liaison, bring forward issues, ideas, and requests from the student body, report committee activities back to students.
- Committee charge includes: (i) recommend policies and plans for student curriculum; (ii) develop and implement systems for curriculum evaluation; (iii) recommend curriculum changes; (iv) develop and recommend policies for continuing education; (v) review and approve all newly-developed courses and areas of specialization at the master's and doctoral level.

IRB & Research Oversight

- Design and implement IRB protocols for clinical and translational studies, coordinate submissions, amendments, and renewals.
- Align consent materials, data-use and governance documentation with institutional policies and regulatory requirements.
- Train and support clinicians and fellows on study design, data queries, and documentation quality for ethical review.

Selected Publications (Author, Co-Author, Data Analyst)

- Bouska A., et al. (2025). Integrative genomic and transcriptomic analysis reveals targetable vulnerabilities in angioimmunoblastic T-cell lymphoma. American Journal of Hematology.
- Lone et al. (2025). Cooperative role of distinctive TP53 and PTEN combined loss in the peripheral T cell lymphoma-GATA3 molecular subgroup. Science Advances
- Amador C., et al. (2025). Comprehensive Characterization of Nodal Peripheral T-Cell Lymphoma, Not Otherwise Specified. American Journal of Clinical Pathology.
- Hu, Z., et al. (2025). SIRT7 facilitates endometrial cancer progression by regulating PTEN stability in an estrogen-dependent manner. Nature Communications.
- Amador, C., et al. (2025). Refining diagnostic subtypes of peripheral T-cell lymphoma using a multiparameter approach. Modern Pathology.
- Lone, W., et al. (2025). High-grade B-cell lymphoma not otherwise specified, with DLBCL gene expression signatures. American Journal of Hematology.
- Puniya B.L., et al. Systems biology identifies drug targets in CD4⁺ T-cell immune disorders. npj Syst. Biol. Appl. 2021.
- Shah AR, et al. Genome-scale metabolic reconstruction of marine microalga N. gaditana. Algal Res. 2017.

Conferences & Invited Talks (Selection)

- ASH-2024 - Genomic Characterization of Novel PTCL-Biological Subtypes Reveal Distinctive Therapeutic Vulnerabilities, San Diego Nov 2024
- Society for Hematopathology / EA4HP Workshop - Houston, 2023: A HemePath Portal for Clinico-Genomic Integration (Invited talk).

- ISMB 2018 - Chicago: Community Metabolic Model of Maize Root System (Oral).
- Nebraska Plant Science Retreat 2019 - Lincoln: Root-Type Specific Metabolic Models of Maize (Keynote).

Professional Memberships & Service

- American Society of Hematology (ASH) - Member
- American Society of Hematopathology - Member
- American Public Health Association - Member
- Manuscript reviewer for PLOS ONE, MedComm, Antibiotics, Bioinformatics, npj Systems Biology
- Journals reviewed ~25

Languages

- English: Professional proficiency (writing/reporting, executive briefings)
- Arabic: Reading/Writing: Advanced; Speaking/Listening: Basic (actively learning; able to ramp quickly in Arabic-speaking environment)

References:

- [Names will be provided upon request](#)

Links:

- To find more details and activities I am involved in on my webpage: <https://arshah18.github.io/>
- Research Activities: <https://scholar.google.com/citations?user=Aje0C8AAAAAJ&hl=en>